

Cyber Security & Digital Forensics

CSE - 4207

Lecture - 08

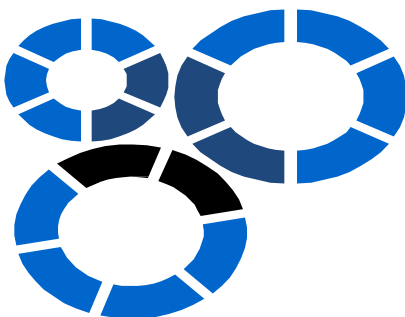
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Professor

Jashore University of Science & Technology

Outline

- Types of malwares
 - Trap Doors
 - Logic Bomb
 - Trojan Horse
- Viruses
 - Boot Sector Virus
 - Macro Virus
 - Rootkit
 - Internet worm
- Malware Detection & Prevention Approaches

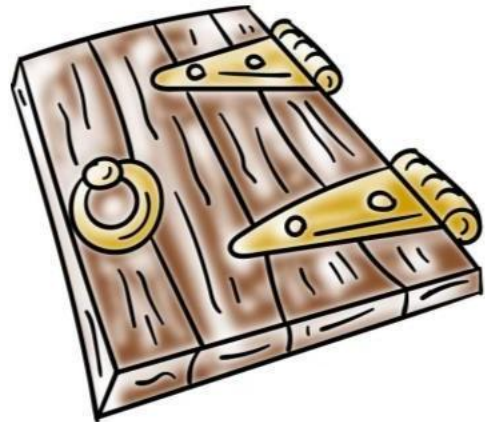


Tyq̇eG o7 fia(ćććoSG So7Cwase efia(wase)

- Needs host program
- Independent

Tyq̇eG o7 fia(ććcoSG So7Cwase efia(wase)

- NeedG ftoGC ąsogsam:



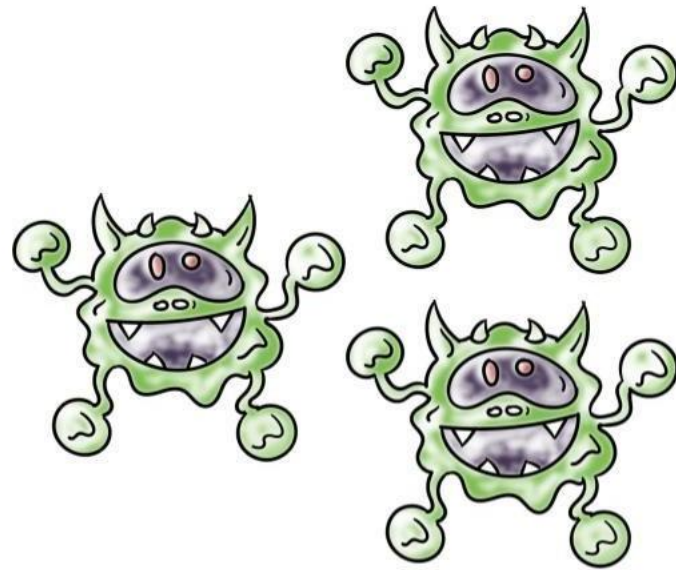
Tsaą doosG



Logćc bombG



Tsojan ftoSGeG

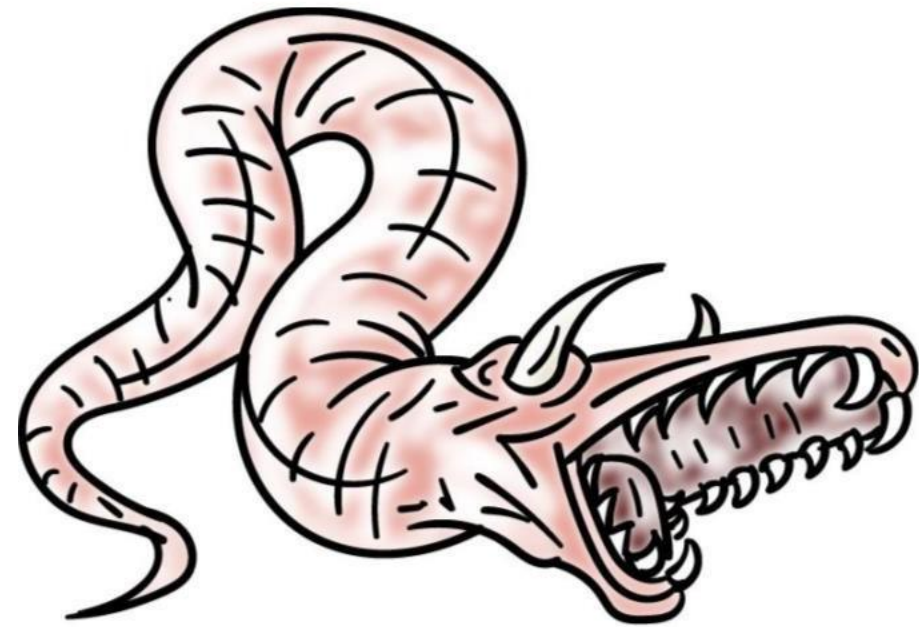


VćsSGeG

BsowGes ą(Sg-ćnG,
exCenGćonG, GcsćąCG

Tyq̃eG o7 fia(ċċoSG So7Cwase efia(wase)

- Indeq̃endenC:



WosmG



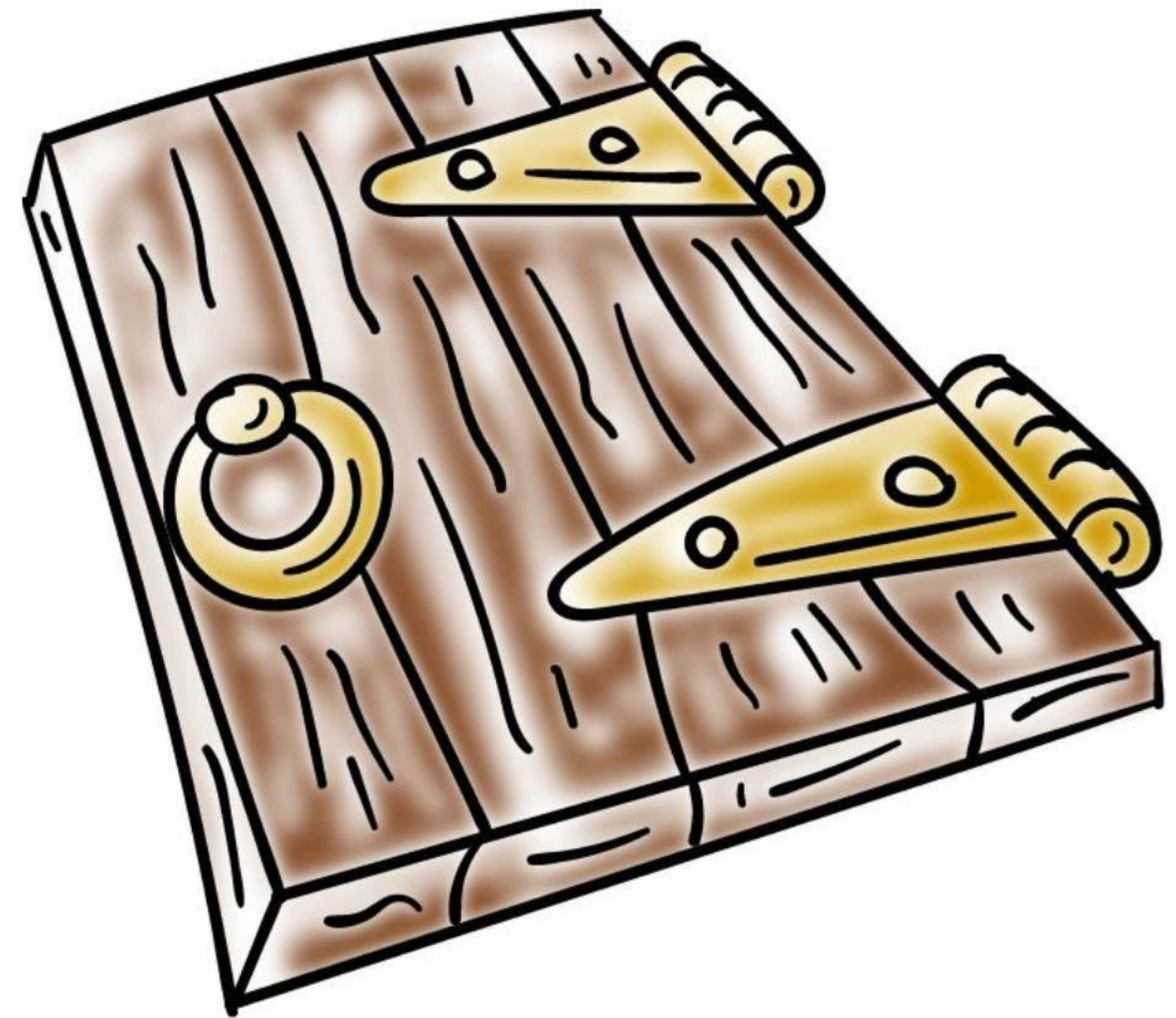
BoCneC



APTG

Tsaã DoosG

- A secret entry point to a program or system.
- Typically works by recognizing some special sequence of inputs or special user ID.



Logic BombG



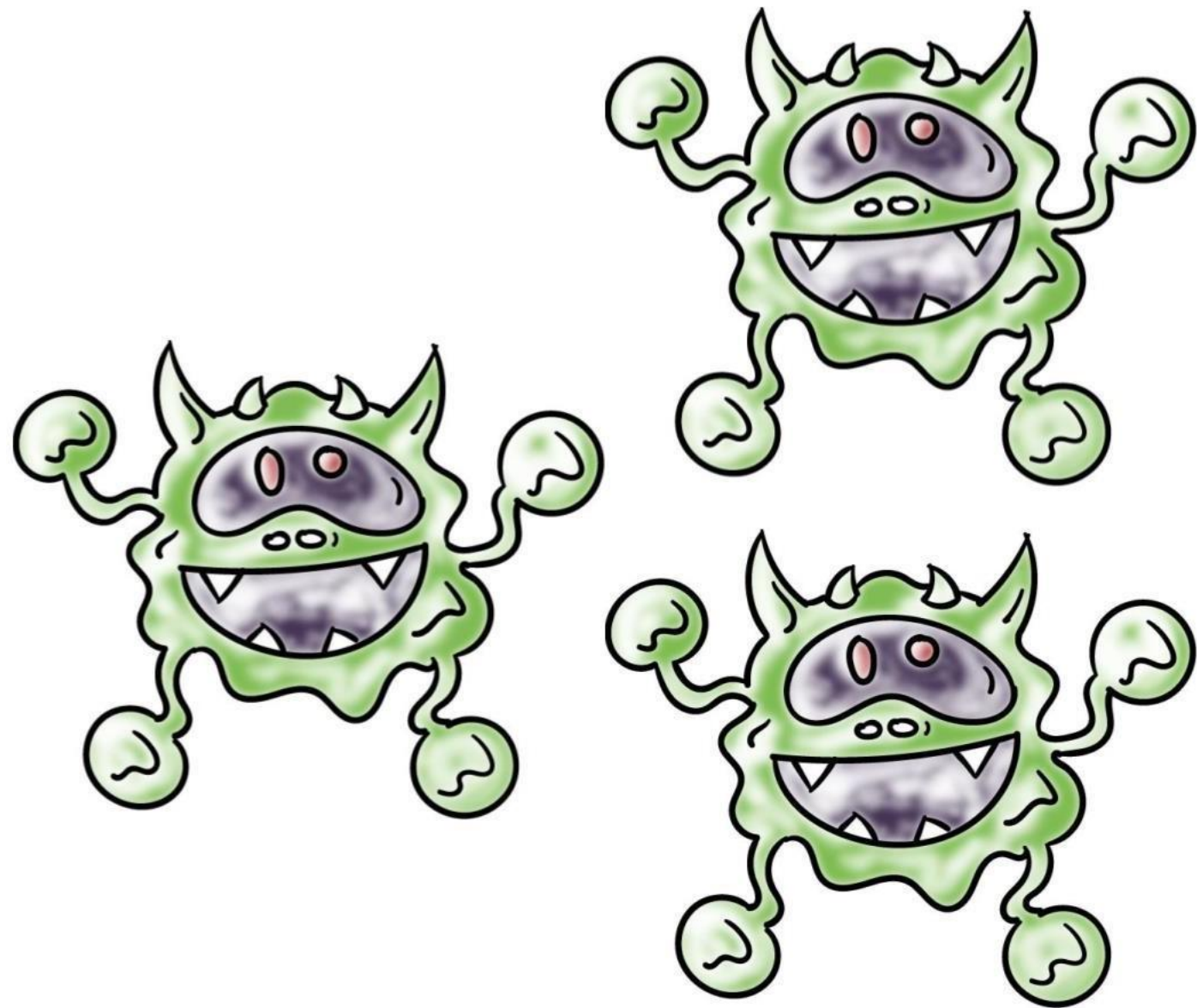
- Embedded in some legitimate Program
- “Explode” or perform malicious activities when certain conditions are met.

Tsojan HosGeG

- Hidden in an apparently useful host program
- Performs some unwanted/harmful function when the host program is executed



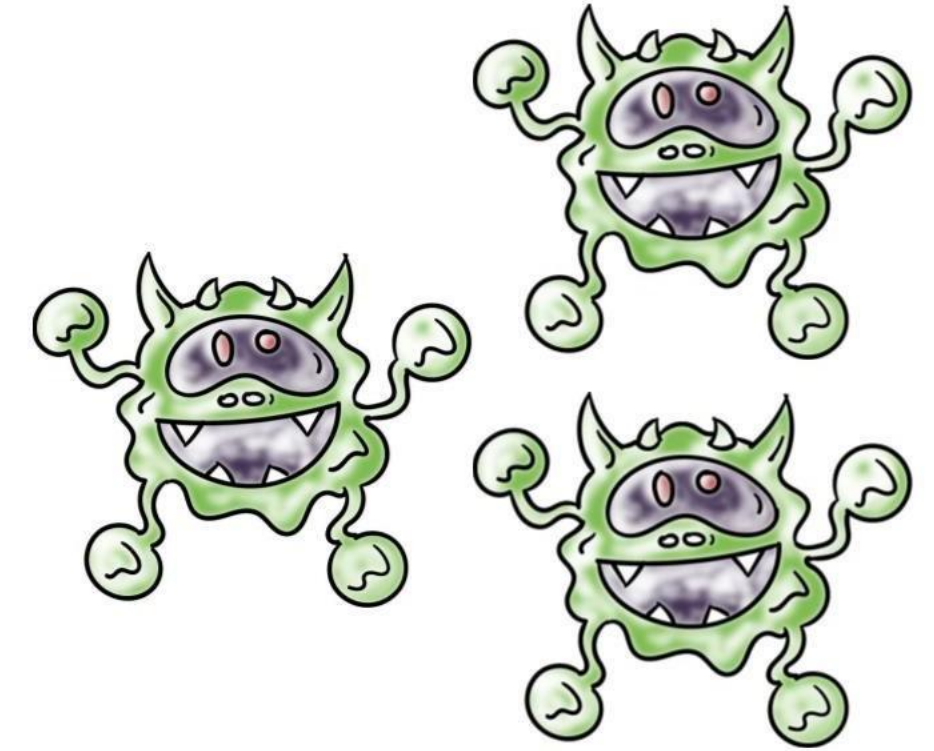
VcsSGeG



- Infect a program by modifying it
- Self-copy into the program to spread

VcsSGeG

FoSs SCageG o7 VcsSGeG



DosmanC

PftaGe

PsoqagaCcon

PftaGe

- E.g. aCCacftmenC
Co emac(

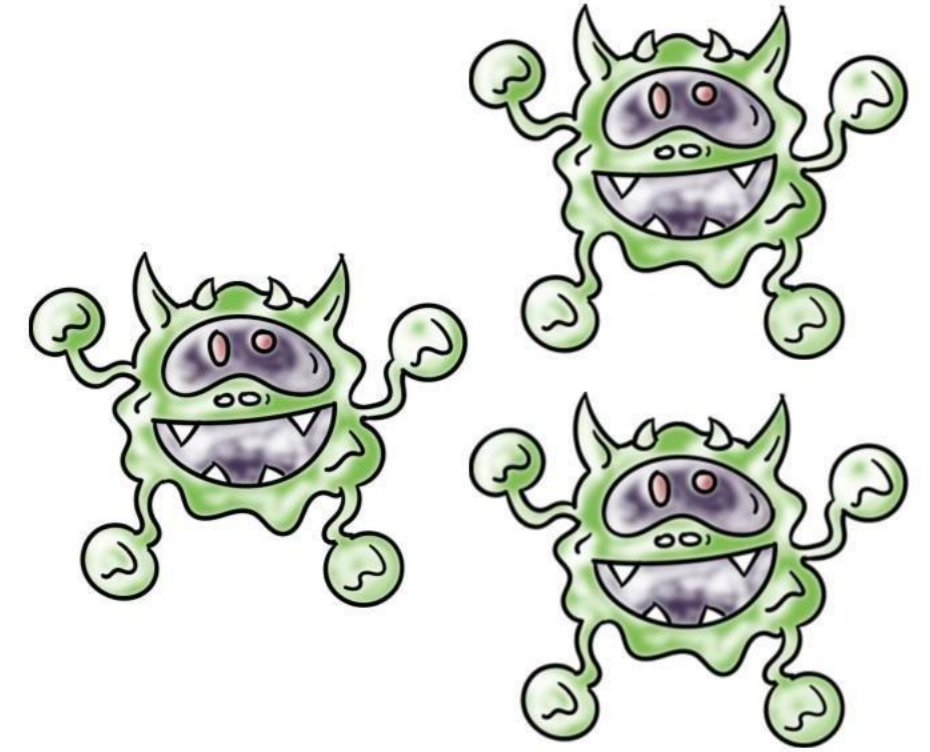
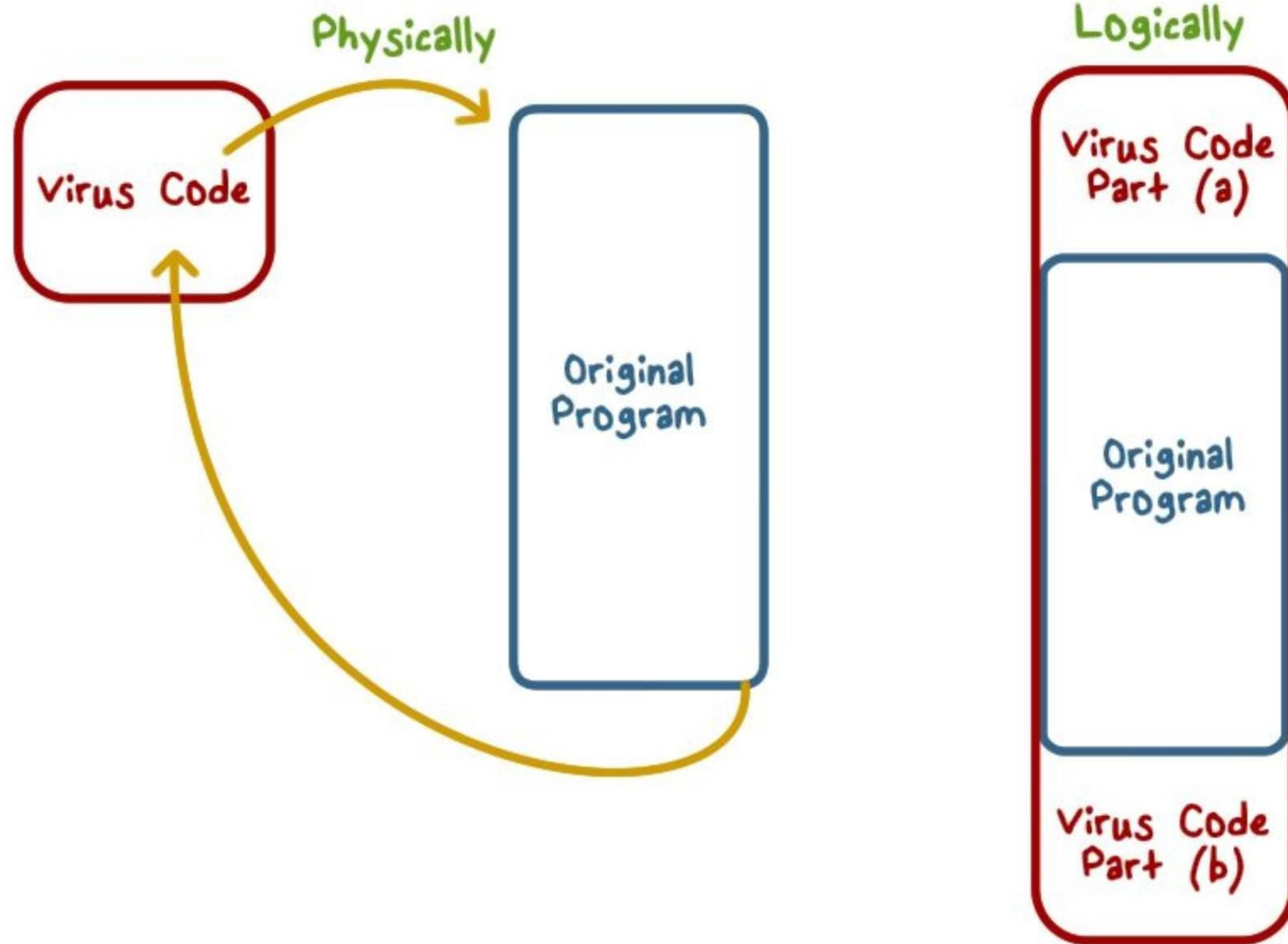
Tscggescng

PftaGe

ExecSCcon

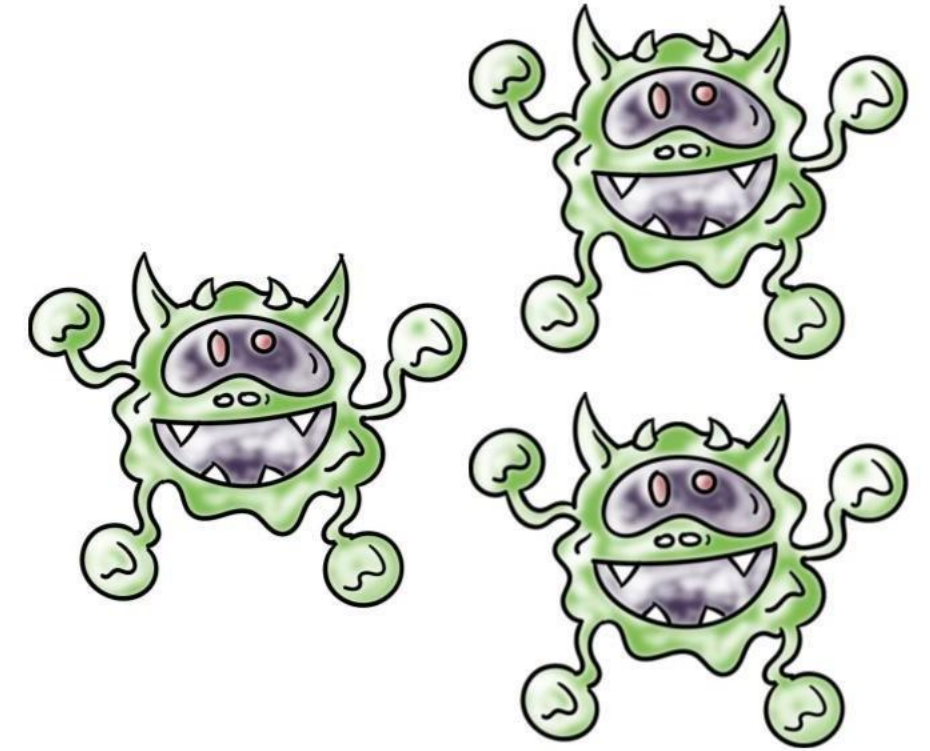
PftaGe

VcsSG SCsScCSse



VcsSG SCsScCSse

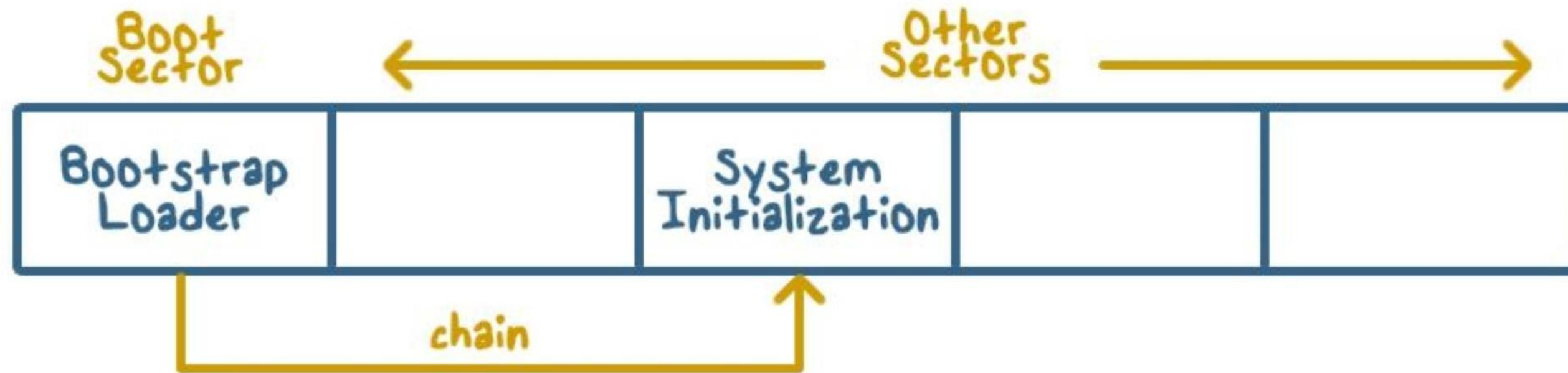
- First line: go to “main” of virus program
- Second line: a special flag (infected or not)
- Main:
 - Find uninfected programs - infect them
 - Do something damaging to the system
 - “Go to” first line of the host program - do normal work
- Avoid detection by looking at size of program
 - Compress/decompress the host program



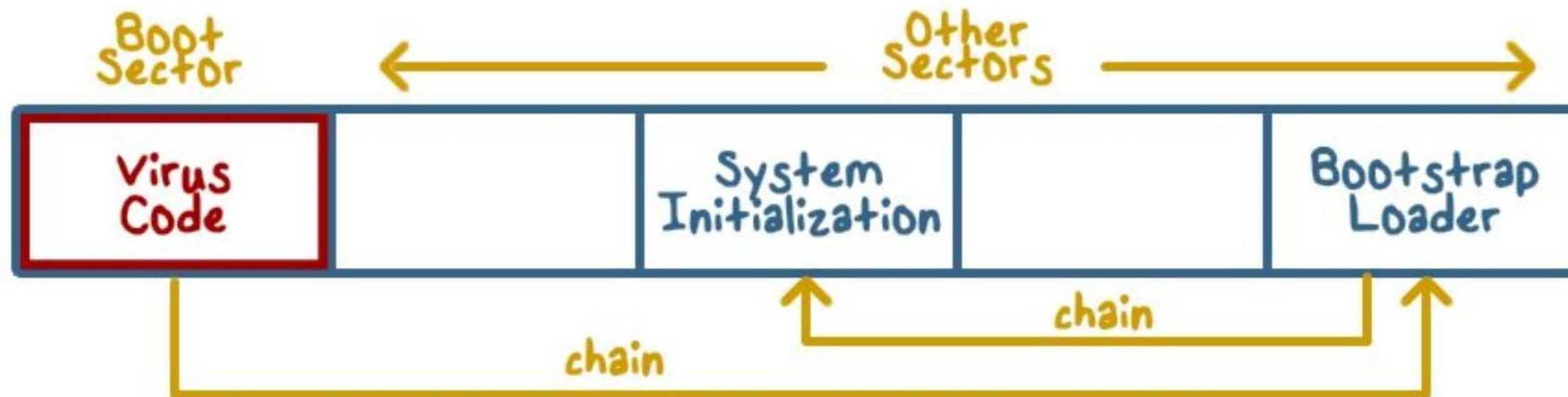
Types of Viruses

- **Parasitic virus:** scan/infect programs
- **Memory-resident virus:** infect running programs
- **Macro virus:** embedded in documents, run/spread when opened
- **Boot sector virus:** run/spread whenever the system is booted
- **Polymorphic virus:** encrypt part of the virus program using a randomly generated key (avoids detection by signature matching)

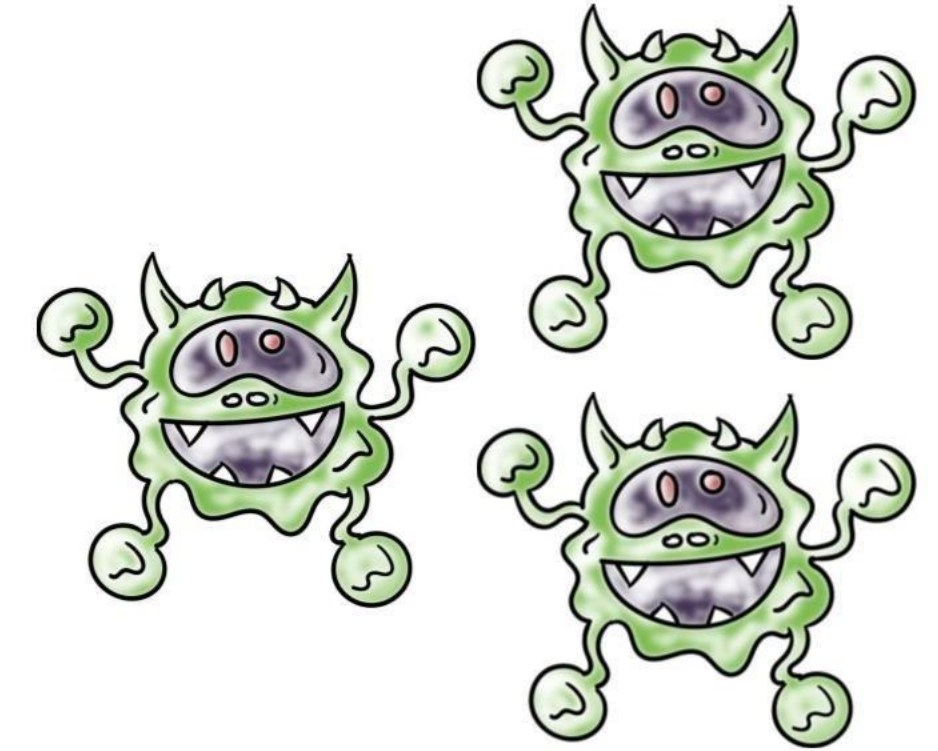
BooC SecCos VcsSG



(a) Before infection



(b) After infection



fiacso VicsGeG

fiacso:

- An executable program (e.g. instructions opening a file, starting an application) embedded in a word processing document, e.g. MS Word



fiacso VcsSGeG

A common CecftncqSe 7os Saseadng:

- A virus macro is attached to a Word Document
- Document is loaded and opened in the host system
- When the macro executes, it copies itself to the global macro file
- The global macro can be activated/spread when new documents are opened



RooCkC

- Resides in operating systems

Modifies OS code and data

Structure

- Helps user-level malware

E.g., hide it from user (not listed in “ls” or “ps” command)



RooCkC

Inspect all files



Windows API

NTQueryDirectoryObject

kernel Native Interface

Device driver functions

Drivers



Example Rootkit structure based on the Dir operation in Windows

RooCkC

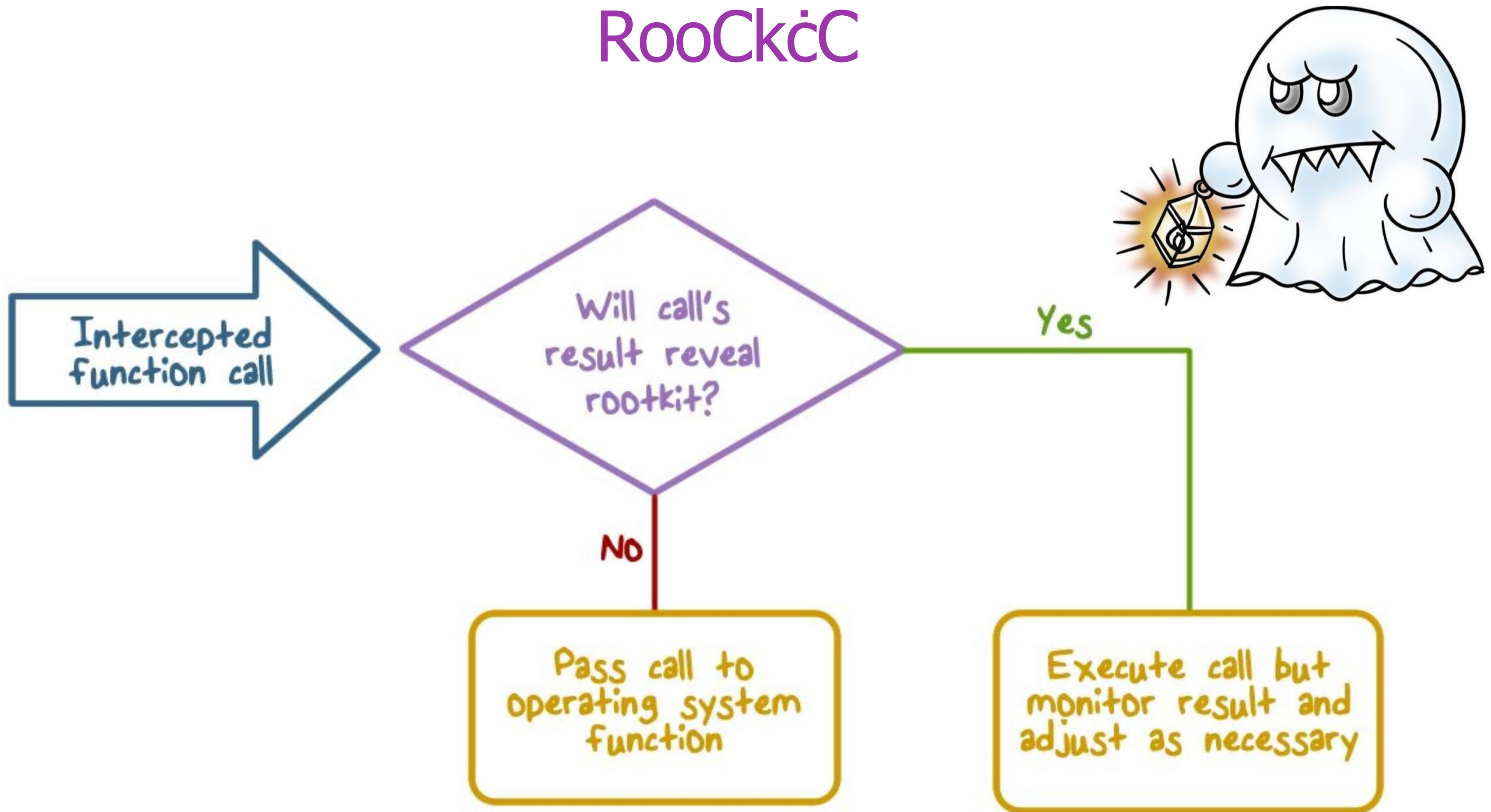


Volume in drive C has no label.
Volume Serial Number is E4C5-A911

Directory of C:\WINNT\APPS

01-09-10	13:34	<DIR>	.
01-09-10	13:34	<DIR>	..
24-07-02	15:00	82,944	CLOCK.AVI
24-07-02	15:00	17,062	Coffee Bean.bmp
24-07-02	15:00	80	EXPLORER.SCF
24-07-08	15:00	256,192	mal_code.exe
22-08-04	01:00	373,744	PTDOS.EXE
21-02-04	01:00	766	PTDOS.ICO
19-06-03	15:05	73,488	regedit.exe
24-07-02	15:00	35,600	TASKMAN.EXE
14-10-02	17:23	126,976	UNINST32.EXE
9 File(s)		966,852 bytes	
2 Dir(s)		13,852,132,800 bytes free	

RooCkC



RooCkC



Inspect all files

```
FindFirstFile()
```

```
{ checkfile  
  FindNextFile()  
  repeat
```

Windows API

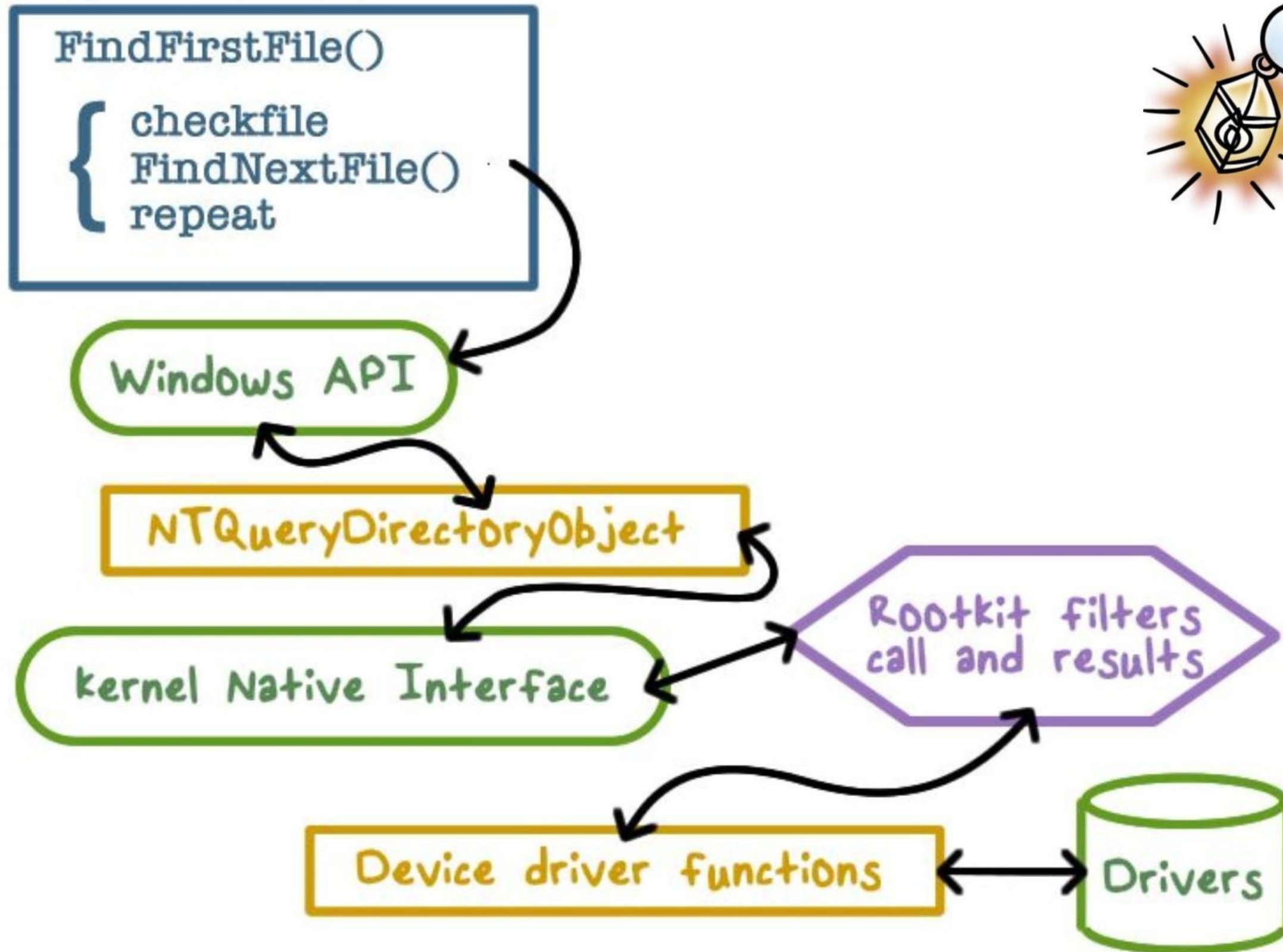
NTQueryDirectoryObject

kernel Native Interface

Rootkit filters
call and results

Device driver functions

Drivers



RooCkC



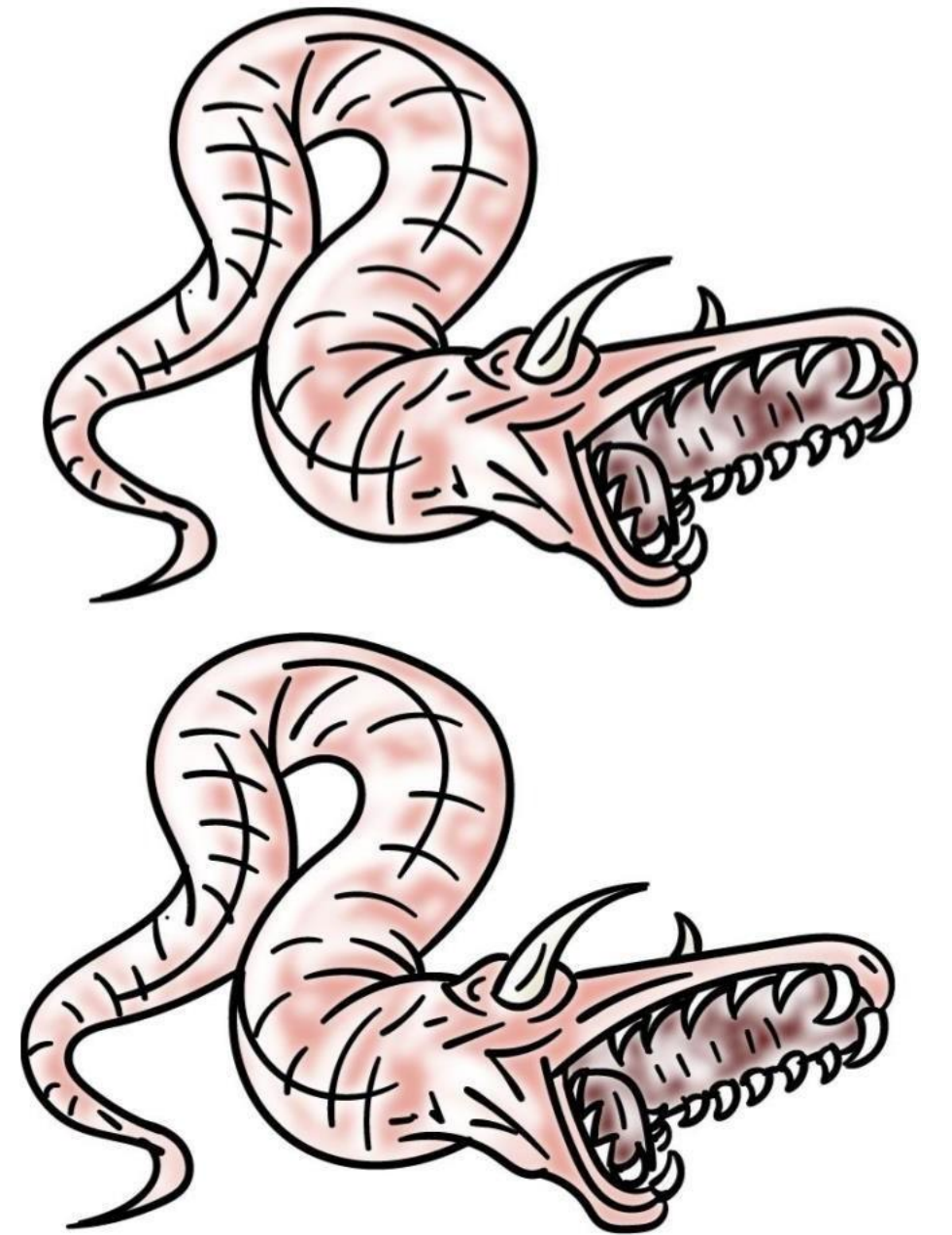
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Directory of C:\WINNT\APPS

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24-07-02	15:00	82,944	CLOCK.AVI
24-07-02	15:00	17,062	Coffee Bean.bmp
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19-06-03	15:05	73,488	regedit.exe
24-07-02	15:00	35,600	TASKMAN.EXE
14-10-02	17:23	126,976	UNINST32.EXE
8 File(s)		710,660 bytes	
2 Dir(s)		13,853,472,768 bytes free	

WosmG

- Use network connections to spread from system to system



Tfte InCesneC Wosm efiosscG Wosm)

WftaC cC dcd:

- Determine where it could spread
- Spread its infection
- Remain undiscovered and undiscoverable



Tfte InCesneC Wosm



E77ecC:

- Resource exhaustion - repeated infection due to programming bug
- Servers are disconnected from the Internet by system admin to stop the infection

Tfte InCesneC Wosm



- Exa(ocC GecSscCy 7(awG
 - Guess password (encrypted passwd file readable)
 - fingerd: buffer overflow
 - sendmail: trapdoor (accepts shell commands)
- Sasead
 - Bootstrap loader to target machine, then fetch
 - rest of code (password authenticated)

Tfte InCesneC Wosm



- Remačn Sn-dčGcovesab(e)
 - Load code in memory, encrypt, remove file
 - Periodically changed name and process ID
- WftaC we (easned:
 - Security scanning and patching
 - Computer Emergency Response Team

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- Prevention: Limit contact to outside world
- Detection and Identification
- Removal



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4 Generations of antivirus software:

- Simple scanners: Use “signatures” of known viruses
- Heuristic scanners: Integrity checking: checksum, encrypted has
- Activity traps
- Full-featured analysis: Host-based, network-based, sandboxing-based



Thank You

